

ex 12

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Run Save Output Input

```
1 clear;
2 clc;
3 close all;
4
5 %% PARAMETERS
6
7 N_subcarriers = 8;           % Number of subcarriers
8 N_subsymbols = 4;           % Number of subsymbols
9 M = 4;                       % QPSK modulation
10 sps = 8;                     % Samples per symbol
11
12 slice_names = {'OFDMA','GFDM'};
13
14 %% =====
15 % GENERATE RANDOM QPSK DATA
16 % =====
17
18 data = randi([0 M-1],N_subcarriers,N_subsymbols);
19
20 % QPSK mapping
21 data_mod = exp(1i*2*pi*data/M);
22
23 %% =====
24 % OFDMA WAVEFORM
25 % =====
26
27 % Total number of samples
28 total_samples = N_subcarriers * ...
29                 N_subsymbols * sps;
30
31 ofdm_signal = zeros(1,total_samples);
32
33 for k = 1:N_subcarriers
34
35     % Get QPSK symbols
36     symbols = real(data_mod(k,:));
37
```

OFDMA AND GFDM WAVEFORM ANALYSIS

Number of Subcarriers : 8
Number of Subsymbols : 4
Modulation : QPSK
Samples per Symbol : 8

Simulation completed successfully.

[Execution complete with exit code 0]

